

Consecutive residues of equal multiplicative order: a proof of the conjecture in OEIS A397683

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Abstract

For an integer $m \geq 2$ let $f(m)$ be the number of x with $1 \leq x < m$ such that x and $x + 1$ are both units modulo m and $\text{ord}_m(x) = \text{ord}_m(x + 1)$. OEIS sequence A397683 (D. S. González May, September 2026) conjectures that for odd $m > 1$, $f(m) = 0$ if and only if m is a power of 3 or a power of 7. We prove this conjecture. The only non-elementary input is S. D. Cohen's theorem that every prime $p > 7$ has a pair of consecutive primitive roots. The rest is an explicit Chinese remainder construction, with the residues 4 modulo 3^a and 3 modulo 7^b chosen so that the multiplicative orders balance. The full argument is formalized in Lean 4 with Mathlib. Cohen's theorem enters as an explicit hypothesis and nothing else is assumed.

1 Introduction

For $m \geq 2$ define

$$f(m) = \#\{1 \leq x < m : \gcd(x, m) = \gcd(x + 1, m) = 1, \text{ord}_m(x) = \text{ord}_m(x + 1)\}.$$

Every unit modulo an even m is odd, so x and $x + 1$ cannot both be units and $f(m) = 0$. The interesting case is odd m . OEIS A397683 lists $a(n) = f(2n - 1)$ and records the following.

Conjecture (A397683). For odd m , $f(m) = 0$ if and only if m is a power of 3 or a power of 7. This was verified for $m < 20000$.

Here “power” means 3^a or 7^b with exponent at least 1, since $f(1) = 1$ by convention. The entry already notes that positivity is preserved under coprime products: if $\gcd(r, s) = 1$, $f(r) > 0$ and $f(s) > 0$, then $f(rs) > 0$. Together with Cohen's theorem this settles every odd m coprime to 21. The open part is every odd m divisible by 3 or 7 that is not a pure power of 3 or of 7. There the coprime-product argument fails, because $f(3^a) = f(7^b) = 0$. We close this case.

Theorem 1. Assume Cohen's theorem (Theorem 2). For every odd $m > 1$, $f(m) = 0$ if and only if $m = 3^a$ or $m = 7^b$ for some $a, b \geq 1$.

Prior work. We checked the following on 9 October 2026 and found no existing proof:

- The OEIS entry and its edit history. The last edit was on 20 September 2026, and the conjecture is still marked open. No other OEIS entry references A397683.
- Google DeepMind's `formal-conjectures` repository, including its issues and pull requests.

- The `the-omega-institute/trureturing` repository, an automated pipeline that proves OEIS conjectures, including its issues and pull requests.
- Recent papers proving OEIS conjectures, namely the “OEIS Open” benchmark and S. Fried’s series. Both predate the entry.
- General web searches.

N. A. Carella’s paper [5], which the entry cites, concerns consecutive primitive roots modulo primes only.

2 Tools

Theorem 2 (Cohen [2, 3]; see also [4]). *For every prime power $q > 7$, the field $\mathbb{F}_q$ contains two consecutive primitive elements. In particular, every prime $p > 7$ has consecutive primitive roots $y, y + 1$ modulo p .*

For $p = 5$ the pair $(2, 3)$ works, because both residues have order 4. We write $E_n(z, k)$ for $z^k \equiv 1 \pmod{n}$, so that $\text{ord}_n(z) \mid k \iff E_n(z, k)$. In particular, $\text{ord}_m(x) = \text{ord}_m(x + 1)$ holds if and only if $E_m(x, k) \iff E_m(x + 1, k)$ for every k .

Lemma 3 (Order profile). *Let p be an odd prime and $p \nmid z$. Let d be the order of z modulo p , and suppose $p^2 \nmid z^d - 1$. Then for every $n \geq 1$ and every k ,*

$$z^k \equiv 1 \pmod{p^n} \iff dp^{n-1} \mid k.$$

Proof. Both sides force $d \mid k$. Write $k = dj$ with $j \geq 1$; the case $j = 0$ is trivial. By the lifting-the-exponent lemma, $v_p((z^d)^j - 1) = v_p(z^d - 1) + v_p(j) = 1 + v_p(j)$. Hence $p^n \mid z^k - 1$ if and only if $p^{n-1} \mid j$, which holds if and only if $dp^{n-1} \mid k$. $\square$

Lemma 4 (Lifting a consecutive pair). *Let $p \geq 5$ be prime, and let $y, y + 1$ be primitive roots modulo p . Then there is $t \in \{0, 1, 2\}$ such that $y' = y + tp$ and $y' + 1$ both satisfy $p^2 \nmid (\cdot)^{p-1} - 1$. Consequently both y' and $y' + 1$ have order $\varphi(p^n) = (p - 1)p^{n-1}$ modulo p^n for every $n \geq 1$.*

Proof. Modulo p^2 , $(y + tp)^{p-1} \equiv y^{p-1} + (p - 1)y^{p-2}tp$. Suppose two values $t_1 \neq t_2$ both give $(y + t_i p)^{p-1} \equiv 1 \pmod{p^2}$. Then $p \mid (p - 1)y^{p-2}(t_1 - t_2)$, so $p \mid t_1 - t_2$. This is impossible, since $|t_1 - t_2| \leq 2 < p$. So at most one t is bad for y , and likewise at most one for $y + 1$, which leaves a good t among the three. The order statement then follows from Lemma 3 with $d = p - 1$. $\square$

Lemma 5 (Profiles modulo 3^a and 7^b). *For $a, b \geq 1$ and all k :*

$$\begin{aligned} 4^k \equiv 1 \pmod{3^a} &\iff 3^{a-1} \mid k, & 5^k \equiv 1 \pmod{3^a} &\iff 2 \cdot 3^{a-1} \mid k, \\ 3^k \equiv 1 \pmod{7^b} &\iff 6 \cdot 7^{b-1} \mid k, & 4^k \equiv 1 \pmod{7^b} &\iff 3 \cdot 7^{b-1} \mid k. \end{aligned}$$

Proof. Apply Lemma 3. The orders modulo the prime are $\text{ord}_3(4) = 1$, $\text{ord}_3(5) = 2$, $\text{ord}_7(3) = 6$ and $\text{ord}_7(4) = 3$. The non-divisibility conditions are $9 \nmid 3$, $9 \nmid 24$, $49 \nmid 728$ and $49 \nmid 63$. $\square$

3 Proof of Theorem 1

Pure powers have $f = 0$.

- $m = 3^a$, $a \geq 1$. If x and $x + 1$ are units modulo 3, then $x \equiv 1 \pmod{3}$. So $x^{3^{a-1}} \equiv 1 \pmod{3^a}$. However, 3^{a-1} is odd, so $(x + 1)^{3^{a-1}} \equiv 2^{3^{a-1}} \equiv 2 \pmod{3}$, and $(x + 1)^{3^{a-1}} \not\equiv 1$. The exponent $k = 3^{a-1}$ separates the two orders.
- $m = 7^b$, $b \geq 1$. Here $x \bmod 7 \in \{1, \dots, 5\}$. The orders modulo 7 of the consecutive pairs $(x, x + 1)$ are

$$(1, 3), (3, 6), (6, 3), (3, 6), (6, 2).$$

In each case there is a $d \in \{1, 3, 2\}$ such that exactly one of $x^d, (x + 1)^d$ is $\equiv 1 \pmod{7}$. Take $k = d \cdot 7^{b-1}$. The side with $z^d \equiv 1 \pmod{7}$ satisfies $z^k \equiv 1 \pmod{7^b}$. On the other side, $7^{b-1} \equiv 1 \pmod{6}$ gives $w^k \equiv w^d \not\equiv 1 \pmod{7}$. Again the orders differ.

Every other odd $m > 1$ has $f(m) > 0$. Write $m = 3^a 7^b r$ with $\gcd(r, 21) = 1$. Since m is not a pure power, either $a, b \geq 1$ or $r > 1$.

The part r . For each $p^n \parallel r$ we have $p \geq 5$ and $p \neq 7$. Theorem 2 (or the pair $(2, 3)$ when $p = 5$) together with Lemma 4 gives y_p with $y_p, y_p + 1$ both of order $\varphi(p^n)$ modulo p^n . Glue these by the CRT into y_r . Then for every k , $E_r(y_r, k) \iff E_r(y_r + 1, k)$, and $E_r(y_r, k)$ forces $2 \mid k$ when $r > 1$, because every $\varphi(p^n)$ is even.

The construction. By the CRT choose x with

$$x \equiv 4 \pmod{3^a}, \quad x \equiv 3 \pmod{7^b}, \quad x \equiv y_r \pmod{r}.$$

Then $x + 1 \equiv 5, 4, y_r + 1$ respectively, so x and $x + 1$ are units modulo m . By Lemma 5, writing $R(k)$ for $E_r(y_r, k)$:

$$\begin{aligned} E_m(x, k) &\iff [a = 0 \vee 3^{a-1} \mid k] \wedge [b = 0 \vee (2 \mid k \wedge 3 \cdot 7^{b-1} \mid k)] \wedge R(k), \\ E_m(x + 1, k) &\iff [a = 0 \vee (2 \mid k \wedge 3^{a-1} \mid k)] \wedge [b = 0 \vee 3 \cdot 7^{b-1} \mid k] \wedge R(k). \end{aligned}$$

Here we used $\gcd(2, 3^{a-1}) = \gcd(2, 3 \cdot 7^{b-1}) = 1$. The two right-hand sides differ only in where the condition $2 \mid k$ appears.

- If $a, b \geq 1$, each side contains the condition $2 \mid k$.
- If exactly one of a, b is positive, or both are 0, then $r > 1$ and $R(k)$ implies $2 \mid k$.

Either way the two sides are equivalent for every k . Hence $\text{ord}_m(x) = \text{ord}_m(x + 1)$, and $f(m) \geq 1$ after reducing x modulo m . $\square$

Remark 6. For $m = 3^a 7^b$ the orders are $\text{ord}_m(x) = \text{lcm}(3^{a-1}, 6 \cdot 7^{b-1})$ and $\text{ord}_m(x + 1) = \text{lcm}(2 \cdot 3^{a-1}, 3 \cdot 7^{b-1})$. Both equal $2 \cdot 3^{\max(a-1, 1)} \cdot 7^{b-1}$. The construction was checked directly for every odd $m \leq 30000$.

4 Lean formalization

The proof is formalized in Lean 4 (v4.33.1) with Mathlib, in about 520 lines; see the accompanying file `A397683.lean`. The function f is defined literally as in the OEIS. Cohen's theorem is the explicit hypothesis

```
def CohenConsecutive : Prop :=
  ∀ p : ℕ, p.Prime → 7 < p →
    ∃ y : ZMod p, orderOf y = p - 1 ∧ orderOf (y + 1) = p - 1
```

and the main theorem reads

```
theorem conjecture (hC : CohenConsecutive) {m : ℕ}
  (hm : 1 < m) (hodd : Odd m) :
  f m = 0 ↔ (∃ a, m = 3 ^ a) ∨ (∃ b, m = 7 ^ b)
```

#print axioms reports only propext, Classical.choice and Quot.sound. In particular there is no sorryAx.

The lifting-the-exponent step uses Mathlib’s `padicValNat.pow_sub_pow`. The lift modulo p^2 uses `sq_dvd_add_pow_sub_sub`. The CRT gluing uses `Nat.chineseRemainder` and `Nat.modEq_and_modEq_iff_modEq_mul`. The reduction over the prime-power factorization uses `Nat.recOnPrimeCoprime`.

Cohen’s theorem itself, which rests on character-sum estimates over finite fields, is not formalized. To our knowledge it is not in Mathlib.

References

- [1] D. S. González May, Sequence A397683, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A397683>, 2026.
- [2] S. D. Cohen, Consecutive primitive roots in a finite field, *Proc. Amer. Math. Soc.* 93 (1985), 189–197.
- [3] S. D. Cohen, Consecutive primitive roots in a finite field II, *Proc. Amer. Math. Soc.* 94 (1985), 605–611.
- [4] S. D. Cohen, T. Oliveira e Silva and T. Trudgian, On consecutive primitive elements in a finite field, *Bull. London Math. Soc.* 47 (2015), 418–426; arXiv:1410.6210.
- [5] N. A. Carella, Least consecutive pair of primitive roots, arXiv:2604.13090, 2026.
- [6] The mathlib Community, The Lean mathematical library, *CPP 2020*.